specific purpose, like images of road signs to be fed to a machine learning model for training.

If you don’t care about the background, road signs themselves use a very limited set of colors, so you can decide it is fine to down sample the colors you use to a 256-color scale[17]: this will allow a factor 3 saving for your storage, which means 4 GBs every thousand pictures stored (if you are storing or processing them on the cloud, this likely means a huge saving of time and money).

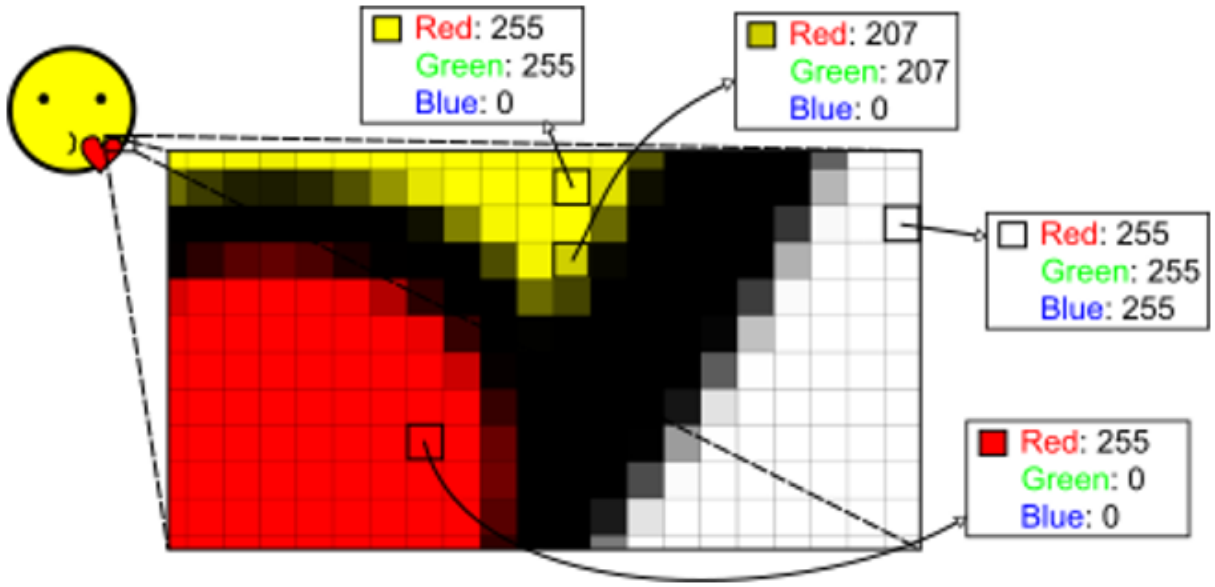


Figure 11.9 RGB bitmap format: an image is made of tiny points, its pixels, and for each of them its color is determined by the combination of three primary colors, red, green and blue, each of which can have an intensity between 0 and 255.

So the problem is, how do you transform each image from one color scale to the other, by keeping the highest fidelity?